

TSAM Assignment 1

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Question 1.

a) Result of running the following commands:

> ping 130.208.246.98

```
kormakur-snorrason@thinkpad:~  
$ ping 130.208.246.98  
PING 130.208.246.98 (130.208.246.98) 56(84) bytes of data.  
^C  
--- 130.208.246.98 ping statistics ---  
51 packets transmitted, 0 received, 100% packet loss, time 51199ms  
+ New Tab
```

> sudo nmap -sS 130.208.246.98

```
kormakur-snorrason@thinkpad:~  
$ sudo nmap -sS 130.208.246.98  
[sudo] password for kormakur-snorrason:  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-19 09:33 GMT  
Nmap scan report for 130.208.246.98  
Host is up (0.0069s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
  
Nmap done: 1 IP address (1 host up) scanned in 4.86 seconds
```

b) Establishing connection and capturing packet:

> ncat connection:

<pre>kormakur-snorrason@thinkpad:~ \$ ncat 130.208.246.98 5100 Remember to check for network connectivity</pre>	<pre>Last login: Wed Aug 26 08:39:03 2026 from 130.208. kormakur24@tsam:~\$ ncat -ln 5100 Remember to check for network connectivity</pre>
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> tcpdump output:

```
kormakur-snorrason@thinkpad:~  
$ sudo tcpdump -vAX -i wlp0s20f3 'port 5100'  
tcpdump: listening on wlp0s20f3, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
Re09:55:32.428679 IP (tos 0x0, ttl 64, id 40513, offset 0, flags [DF], proto TCP (6), length 95)  
  thinkpad.48614 > 130.208.246.98.5100: Flags [P.], cksum 0x8baee (incorrect -> 0xd5b8), seq 1732830826:1732830869, ack 162824696, win 63, options [nop,nop,TS val 914168365 ecr 2769021023],  
  length 43  
    0x0000: 4500 005f 9e41 4000 4006 11b5 0a64 0706  E...A@.@....d..  
    0x0010: 82d0 f662 bde6 13ec 6748 e66a 09b4 81f8  ..b...gh.j...  
    0x0020: 8018 003f 8aee 0000 0101 080a 367d 1a2d  ...?.....6)..  
    0x0030: a50b e85f 5265 6d65 6d62 6572 2074 6f20  ... Remember.to.  
    0x0040: 6368 6563 6b20 666f 7220 6e65 7477 6f72  check.for.networ  
    0x0050: 6b20 636f 6e6e 6563 7469 7669 7479 0a    k.connectivity.  
09:55:32.437098 IP (tos 0x0, ttl 58, id 38164, offset 0, flags [DF], proto TCP (6), length 52)  
  130.208.246.98.5100 > thinkpad.48614: Flags [.], cksum 0x0f05 (correct), ack 43, win 509, options [nop,nop,TS val 2769041930 ecr 914168365], length 0  
    0x0000: 4500 0034 9514 4000 3a06 2113 82d0 f662  E..4..@.:!....b  
    0x0010: 0a64 0706 13ec bde6 09b4 81f8 6748 e695  .d.....gh..  
    0x0020: 8018 01fd 0f05 0000 0101 080a a50c 3a0a  .....  
    0x0030: 367d 1a2d 6j..
```

- c) In your own words, explain what each of the three commands does.
- ping:** It uses the ICMP protocol's requests to receive a response from a host.
- nmap:** It scans a network and lists every visible host on said network.
- ncat:** It reads and writes data across networks from the command line.
- tcpdump:** It intercepts network traffic and dumps the packets to the terminal.
- d) Repeat (b) above, but use UDP instead of TCP.

```
kormakur-snorrason@thinkpad:~$ ncat -u 130.208.246.98 5100
Remember to check for network connectivity
Remember to check for network connectivity

kormakur24@tsan:~$ man ncat
kormakur24@tsan:~$ ncat -u -ln 5100
Remember to check for network connectivity
Remember to check for network connectivity

kormakur-snorrason@thinkpad:~$ sudo tcpdump -vAX -i wlp0s20f3 -n udp 'port 5100'
tcpdump: listening on wlp0s20f3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:31:05.657311 IP (tos 0x0, ttl 64, id 60994, offset 0, flags [DF], proto UDP (17), length 71)
10.100.7.6.36772 > 130.208.246.98.5100: UDP, length 43
0x0000: 4500 0047 ee42 4000 4011 c1c6 0a64 0706 E..G.B@.....d..
0x0010: 82d0 f662 8fa4 13ec 0033 8ae1 5265 6d65 ...b.....3...Reme
0x0020: 6d62 6572 2074 6f20 6368 6563 6b20 666f mber.to.check.fo
0x0030: 7220 6e65 7477 6f72 6b20 636f 6e6e 6563 r.network.connec
0x0040: 7469 7669 7479 0a          tivity.
```

- e) (5 points) How does the tcpdump output differ when you use UDP vs TCP? Why?
- When using TCP the data packet is a lot larger because of the size difference between the headers (52 vs 8 bytes). On top of that, UDP is connectionless which means that data packets are fired with no acknowledgement on delivery. Contrast that with TCP where delivery is acknowledged and a response packet is sent back from the server.

Question 2.